ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE-AI)

Semester: II

SUBJECT CODE: BCSAI203

DESIGN THINKING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is Human-Centered Design? Give an example.

(ii) Name two challenges in adopting AI chatbots in customer service.

(iii) How does Empathy Mapping help in improving travel booking apps?

(iv) Why is Learning by Doing essential for engineers?

(v) Why is User Feedback important in social media platform updates?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) List three creativity skills used in product design.

(ii) How does Qualitative Research help in improving smart home technology?

(iii) Define Latent Needs with an example from online shopping.

(iv) Why is Prototyping useful before launching a new gadget?

(v) How does Innovation Diffusion apply to self-driving cars?

SECTION – B

NOTE: Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) A food delivery app wants to improve customer retention. How can Empathy and User Research help?

OR

(b) A university wants to improve online learning for students. How can Problem Reframing help?

(ii)(a) Your team is designing a new smart watch. Suggest three Idea Generation techniques for innovation.

OR

(b) A supermarket wants to introduce AI powered-checkout systems. How can Creativity in Teaching help customers to use it easily?

Q.4.(i)(a) A company is launching a voice assistant for senior citizens. How can Prototyping and Testing make it better?

OR

(b) A city wants to reduce traffic congestion. How can Human-Centred Design help?

(ii)(a) A fitness app is struggling to attract users. Explain how Innovation Diffusion can help it grow.

OR

(b) How did companies like Amazon or Flipkart use Experimentation in Design Thinking to enhance user experience?

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) A hospital is creating a voice-based appointment booking system. Explain the Research Design Process needed.

OR

(b) Explain Creativity vs. Innovation with examples from companies like Apple or Tesla.

Q.6.(a) A rural bank wants to introduce cashless payments. How would you Observe, Describe, and Participate to understand user needs?

OR

(b) How can Design Thinking principles improve customer experience in public transport ticket booking system?